

Assignment 2: Multilevel Models for Educational Data

ADVANCED STATISTICAL LEARNING AND MODELING - MODULE B / Giulia Vannucci

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Instructions

This assignment involves the application of mixed models and multilevel modeling techniques to educational data. Students are free to use either R or Python to complete the tasks.

Assignments may be completed individually or in groups. However, each student must submit their own version of the assignment by uploading it to the following directory on Microsoft Teams:

Assignments → **Assignment2** → [create a folder named: `name_surname_assignment2`]

Each student's folder must contain:

- A PDF report including the exercise;
- The code used for the analysis (in '.R', '.py', or '.ipynb' format);
- A note stating whether the work was done individually or as part of a group, and if in a group, the names of all group members.

Deadline: the day of the end of exam inscription. While there is no penalty for late submission, students are strongly encouraged to complete the assignment as early as possible to allow time for feedback and clarification.

Exercise: Multilevel Modeling of Standardized Math Scores

Dataset

The dataset `INVALSI_data_MAT_2324` contains standardized test scores in mathematics from the INVALSI national assessment for fifth-grade (G05) students in Italy. The data come from a two-stage stratified probability sample, where:

- First-stage units are school campuses (plessi scolastici);
- Second-stage units are class sections.

The testing is conducted under controlled conditions with external observers.

Additional information about variable definitions is provided in the accompanying file:

`MAT_05-2023-24_datadescription`.

Objective

The objective of this exercise is to model math performance of students by applying appropriate **mixed-effects models** that account for the hierarchical structure of the data (e.g., students nested within classes or schools).

Instructions

1. Data Preparation

- Import the dataset and inspect the variable types and completeness.
- Exclude missing or unusable data, and clearly motivate these choices in your report.
- I suggest the following variables for the analysis: “gender”, “month”, “year”, “origin”, “nursery_attendance”, “kindergarten_attendance”, “father_education”, “mother_education”, “father_occupation”, “mother_occupation”, “Region_Code”, “GeoArea_5”, “ESCS_student”, “ESCS_class”, “WLE_MATH_200”, “math_score_raw”, “math_score_percentile”. Feel free to include additional variables if relevant, and clearly motivate your choices.

Response Variable

Choose one among the following as your response variable (AND DO NOT USE THE OTHER OPTIONS AS PREDICTORS): - WLE_MATH_200 - math_score_raw - math_score_percentile

If you consider it appropriate, you may transform your chosen response variable (e.g., into a categorical form) and use a GLMM suitable for the transformed outcome.

Justify your choice of response variable.

Variable Classification

For each of the remaining variables, indicate whether it is:

- an individual-level variable;
- a school-level variable;
- a contextual variable.

Modeling

Specify and fit a multilevel (mixed-effects) model appropriate for your response variable and data structure.

- Justify your model structure (e.g., random intercepts, random slopes).
- Report and interpret key parameter estimates.
- Discuss model fit and any diagnostics performed.

Report

Include in your report:

- Preprocessing steps and rationale for variable selection;
- Definition of model hierarchy and structure;
- Estimation results and interpretation of fixed and random effects;
- Diagnostics, goodness-of-fit measures, and model comparison if applicable;
- A brief discussion of your findings in context of educational outcomes.

Submission Checklist

In your personal folder on Teams, include:

- **assignment2_report.pdf** (your final report);
- The script or notebook used (.R, .py, or .ipynb);
- A note specifying individual or group work and, if in a group, the names of all members.

Folder structure:

Assignments → Assignment2 → name_surname_assignment2

Good luck, and don't hesitate to reach out with questions during your work on the assignment!